ENTHUSE, LEADER & ACHIEVER COURSE

PHASE : ALL SRG ENTHUSE, LEADER, ACHIEVER COURSE & LAKSHYA PHASE

TARGET : PRE-MEDICAL 2026

Test Type : **SRG-MAJOR** Test Pattern : **NEET (UG)**

TEST DATE : 10-06-2026

ANSWER KEY

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A.	3	2	2	3	1	4	3	2	1	2	2	4	4	2	2	2	3	1	3	2	2	4	1	3	4	1	1	3	2	3
Q.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
A.	2	1	4	3	3	4	1	1	3	2	1	3	4	4	3	2	4	1	2	1	1	2	1	1	3	2	4	4	2	2
Q.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
A.	2	3	2	4	3	1	1	2	4	4	2	3	1	1	1	2	3	4	2	3	3	4	3	2	3	4	4	2	3	2
Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
A.	3	1	2	2	2	3	1	2	4	1	2	1	4	4	4	3	3	3	2	2	2	2	3	2	1	2	1	3	1	2
Q.	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	3	3	3	1	3	4	1	3	1	4	3	1	3	3	1	3	3	1	2	3	4	4	2	3	1	3	3	2	2	1
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
A.	1	1	4	2	2	2	1	4	3	1	2	3	4	1	3	2	1	3	4	2	2	2	2	4	1	4	4	2	2	2

HINT – SHEET

1. Ans (3)

Angle between two bonds

$$\theta = \cos^{-1}\left[\frac{\left(\hat{i}+\hat{j}+\hat{k}\right)\cdot\left(\hat{i}-\hat{j}-\hat{k}\right)}{\sqrt{1^2+1^2+1^2}\sqrt{1^2+(-1)^2+(-1)^2}}\right]$$

$$= \cos^{-1}\left(-\frac{1}{3}\right)$$

2. Ans (2)

TME = K + U

$= 300.0 \pm 3.0$

$$\% \text{ error} = \frac{3.0}{300.0} \times 100$$

Answer: The percentage error in the measurement of mechanical energy is 1%.

3. Ans (2)

$$\frac{1}{(Ceq)} = \frac{1}{C_1} + \frac{1}{C_2}$$

Differentiate

$$\frac{\Delta Ceq}{(Ceq)^2} = \frac{\Delta C_1}{C_1^2} + \frac{\Delta C_2}{C_2^2}$$

$$Ceq = \frac{C_1C_2}{C_1+C_2} = 1200\,Pf$$

$$\frac{1}{1200}\left[\frac{\Delta Ceq}{Ceq}\right] = \left[\frac{10}{(2000)^2} + \frac{15}{(3000)^2}\right]$$

$$\frac{\Delta Ceq}{Ceq} = 5 \times 10^{-3}$$

$$U = \frac{1}{2} Ceq\, V^2$$

$$\frac{\Delta U}{U} \times 100 = \frac{\Delta C_{eq}}{C_{eq}} \times 100 + \frac{2\Delta V}{V} \times 100$$

$$= \frac{1}{200} \times 100 + 2 \times \frac{0.02}{5} \times 100 = 1.30\,\%$$

The percentage error in the calculation of the energy stored in this combination of capacitors is 1.3%.

4. Ans (3)

The least count is the pitch divided by the number of circular divisions. The diameter is the sum of the main scale reading and the circular scale reading, corrected for the zero error.

MSR = 2.5 mm

$CSR = 45 \times \frac{0.5}{50}$ mm

= 0.45 mm

Diameter reading = MSR + CSR − zero error

= 2.5 + 0.45 − (−0.03)

= 2.98 mm

5. Ans (1)

$\text{Figure of Merit} = C = \frac{i}{\theta}$

$= C = \frac{6 \times 10^{-3}}{2} = 3 \times 10^{-3}$ A/div.

6. Ans (4)

Smaller the least count greater the precision.

$\text{Mean absolute error } |\Delta x| = \frac{\sum |x - x_i|}{N}$.

7. Ans (3)

It is case of free fall in which initial velocity is zero

$h = ut + \frac{1}{2}at^2$

$h = \frac{1}{2}gt^2$

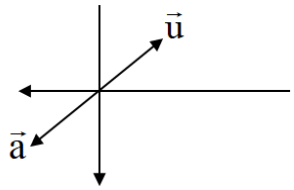
$h \propto t^2$

$t \propto \sqrt{H}$

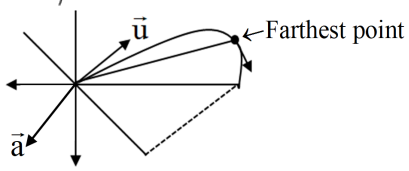
If we require regular time interval so h must be very so option (c) is correct.

8. Ans (2)

If $\vec{u} \| (-\vec{a})$ where $\vec{a} = \vec{a}_x + \vec{a}_y$ Then path is straight line (D) is true



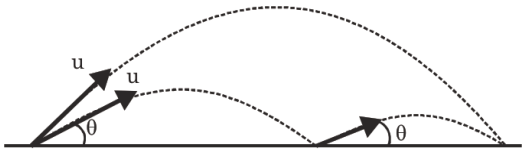
If $\vec{u} \not\| \vec{a}$ then path is parabolic (C) is true



(A) At farthest point velocity is not horizontal.

(B) Vertical motion is unaffected by horizontal resistance ⟶ time up = time down ⟶ True.

9. Ans (1)



$D = \frac{u^2 \sin[2 \times 45°]}{g} = \frac{u^2 \sin 2\theta}{g} + \frac{u^2 \sin 2\theta}{4g}$

$1 = \frac{5}{4}\sin 2\theta \Rightarrow \sin 2\theta = \frac{4}{5}$

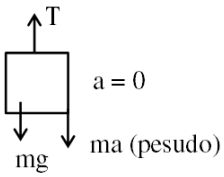
$2\theta = 53° \Rightarrow \theta = 26.5°$

$\frac{T_1}{T_2} = \frac{\frac{2u \sin 45°}{g}}{\frac{2u \sin\theta}{g} + 2\left(\frac{u}{2}\right)\frac{\sin\theta}{g}} = \frac{2\left(\frac{1}{\sqrt{2}}\right)}{3\sin(26.5°)} = \frac{\sqrt{10}}{3}$

10. Ans (2)

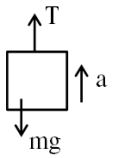
Real forces are frame independent. So value of tension in string will come out same wrt both observer.

FBD wrt A



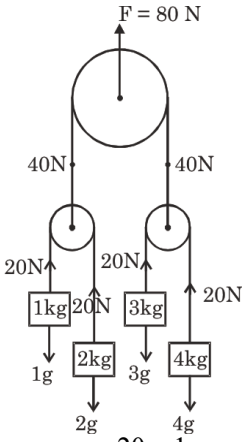
T=m (g+a)

FBD wrt B



T=m (g+a)

11. Ans (2)



$a(1kg) = \frac{20 - 1g}{1} = 10 m/s^2 (\uparrow)$

$a(2kg) = 0$

$a(3kg) = \frac{3g - 20}{3} = \frac{10}{3} m/s^2 (\downarrow)$

$a(4kg) = \frac{4g - 20}{4} = 5\ m/s^2 (\downarrow)$

12. Ans (4)

$0 = 4 - (\mu g) \times 4$ ... (i)

$\mu = 0.1$

In first 2 sec

$s_2 = 4 \times 2 - \frac{1}{2} \times 1 \times 4$

$s_2 = 6$ m ... (ii)

In first 4 sec

$s4 = 4 \times 4 - \frac{1}{2} \times 1 \times 16 = 8m$

In next 2 sec

$s_4 - s_2 = 2m$ ... (iii)

By motion under gravity analogy

$\frac{t_1}{t_2} = \frac{\left(\sqrt{2} - \sqrt{1}\right)}{1}$, where t_1 & t_2 are time taken to cover first half and second half.

13. Ans (4)

For $0 < F < 40$, there is no slipping anywhere. Between 6kg and ground slipping starts for $F \geq 40$. Between 2 blocks slipping starts for $F \geq 72$N so

for $F = 36$N, $f_1 = 0$, $f_2 = 36$N

$F = 48$N, $f_1 = 4$N, $f_2 = 32$N

$F = 64$N, $f_1 = 8$N, $f_2 = 32$N

$F = 96$N, $f_1 = 8$N, $f_2 = 32$N

14. Ans (2)

To reach B, first it has to cross 100 m height so KE must be at least = mg(100) = 1000 J

Final Answer: The correct option is (2)

15. Ans (2)

Solution: The LED glows (Output Y = 1) only when current flows through it. The switches A and B are in parallel across the LED and its series resistor R. Closing a switch shorts the LED, so no current flows through it.

A	B	Y (LED)
0 (Open)	0 (Open)	1 (Current flows through LED)
0 (Open)	1 (Closed)	0 (Current bypasses LED via B)
1 (Closed)	0 (Open)	0 (Current bypasses LED via A)
1 (Closed)	1 (Closed)	0 (Current bypasses LED via A & B)

The output is HIGH (1) only when both inputs are LOW (0).

This is the characteristic of a NOR gate.

16. Ans (2)

$\left(\text{Number of hole}\right)_{\text{Valence band}} > \left(\text{Number of electron}\right)_{\text{Conduction band}}$

$N_h >> N_e$

So, p–type semiconductor

17. Ans (3)

As RC time constant of the capacitor is quite large ($\tau = RC = 10 \times 10^3 \times 10 \times 10^{-6} = 0.1$sec), it will not discharge appreciably. Hence voltage remains nearly constant.

18. Ans (1)

Let the velocity be v

$\therefore$ Photoelectron follow the parabolic path

$d = \frac{1}{2}at^2$

$\therefore$ Acceleration $a = \frac{eE}{m_e}$ where e and me are change of mass of e^-

$\therefore d = \frac{1}{2}\frac{eE}{m_e}t^2$

$t = \sqrt{\frac{2dm_e}{eE}}$

$\therefore \ell = vt$

$\ell = V\sqrt{\frac{2dm_e}{eE}}$

$\frac{1}{2}mv^2 = \frac{E\ell^2}{4d}$

19. Ans (3)

According to Bohr's theory

$\text{Energy } E_n = \frac{me^4}{8\epsilon_0^2 h^2} \cdot \frac{1}{n^2}$

$\therefore E \propto m$ if $n = 1$

using reduced mass

$m' = \frac{m_1 \times m_2}{m_1 + m_2} = \frac{207m_e \times 1836\, m_e}{207m_e + 1836\, m_e}$

$= m' = 186m_e$

$\frac{E_1}{E_2} = \frac{m_1}{m_2}$

$\frac{13.6}{E_2} = \frac{m_e}{186\, m_e}$ $\therefore E_2 = 2.53 \times 10^3$ eV

$\therefore$ Ionisation potential = 2.53 kV

20. Ans (2)

Question Explanation:

We need to find the mass of fuel consumed.

Concept:

Nuclear physics.

Solution:

$E = \Delta m\, C^2$

$10^9 \times 3600 = \Delta m \times (3 \times 10^8)^2$

$\Delta m = \dfrac{36 \times 10^{11}}{9 \times 10^{16}} = 4 \times 10^{-5}\text{kg}$

$\Delta m = 4 \times 10^{-2}$ gm

Final answer: option(2)

21. Ans (2)

Resolving power $\propto \dfrac{1}{\lambda}$

Since wavelength of electron is much less than visible light, its resolving power will be much more.

22. Ans (4)

Theory notes

23. Ans (1)

Using calorimetery principal

$150 \times 540 + 150 \times 1 \times (100 - T)$

$= (250 + 50) \times 1 \times (T - (-10)) + 800 \times \dfrac{1}{2} \times (0 - (-10))$

$+ 800 \times 80 + 800 \times \dfrac{1}{2} \times (T - 0°)$

By solving it t = 20°C hence steam and ice both become water so amount of water = 150 + 800

24. Ans (3)

Factual

Type of gases	No. of degrees of freedom
Monoatomic gas	3 T
Diatomic + rigid	3T + 2R
Diatomic + non-rigid	3T+ 2R + 1V
Polyatomic	3T + 3R + More than 1V

T = Translational degree of freedom

R = Rotational degree of freedom

V = Vibrational degree of freedom

25. Ans (4)

$A = x_{max} = 2m \quad a_{max} = \omega^2 A$

$P_{max} = m\omega A \quad k_{max} = \dfrac{(8\pi)^2}{2 \times 2}$

$8\pi = 2 \times \pi \times 2$

26. Ans (1)

K and m are not known.

27. Ans (1)

$\text{end correction } e = \dfrac{3\ell_3 - 5\ell_2}{2}$

$e = \dfrac{3 \times 74 - 5 \times 44}{2}$

$e = 1$ cm

OR

$\dfrac{\lambda}{2} = 74 - 44 = 30\,\text{cm}$

$\lambda = 60$ cm

$\ell_1 = \dfrac{\lambda}{4} - e = 15 - e$

$\ell_2 = \dfrac{3\lambda}{4} - e = 45 - e,$

$\ell_2 = 44\text{cm}, \quad e = 45 - 44 = 1\text{cm}$

28. Ans (3)

V at gen "y" from bottom $= \sqrt{\dfrac{\lambda g y}{\lambda}} = \sqrt{gy}$

$\Rightarrow \quad V^2 = O^2 + 2\dfrac{g}{2} y$

$\Rightarrow \quad a = \dfrac{g}{2}\,(\text{cons tan t})$

29. Ans (2)

Material which is most ductile is easy to expand is used for making wire..

30. Ans (3)

Material which breaks just after proportional point.

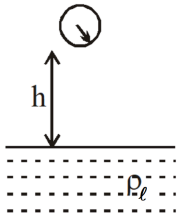
31. Ans (2)

After falling through h, the velocity be equal to terminal velocity

$$\sqrt{2gh} = \frac{2}{9}\frac{r^2 g}{\eta}(\rho_\ell - \rho)$$

$\Rightarrow h = \dfrac{2}{81}\dfrac{r^4 g(\rho_\ell - \rho)^2}{\eta^2}$

$\Rightarrow h \propto r^4$



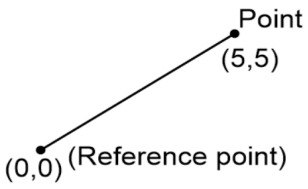
32. Ans (1)

Angular Momentum about a point is given by

Formula $\vec{L} = \vec{r} \times \vec{P}$

$\vec{r}$ = Position rector of Particle w.r.t. reference point

$\vec{P}$ = momentum of particle.



$\vec{v} = 8\hat{i} + 4\hat{j}$

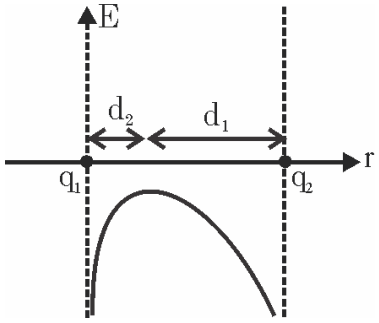
$m = 1\,kg$

Given: $m = 1\ \text{kg}, \vec{r} = 5\hat{i} + 5\hat{j}, \vec{v} = 8\hat{i} + 4\hat{j}$

$$\vec{L} = m(x\hat{i} + y\hat{j}) \times (v_x\hat{i} + v_y\hat{j})$$
$$\vec{L} = m(xv_y - yv_x)\hat{k}$$
$$\vec{L} = 1 \times (5 \times 4 - 5 \times 8)\hat{k}$$
$$\vec{L} = (20 - 40)\hat{k} = -20\hat{k}$$

Final Answer: Option (1)

33. Ans (4)



q_1 is $-$ve

q_2 is +ve

$d_1 > d_2$

So $|q_2| > |q_1|$

34. Ans (3)

equipotential lines are straight and parallel, the potential function must be linear.

V(x, y) = Ax + By + C ____(*)

V(1, 0) = A + C = 10 ___(1)

V(2, 0) = 2A + C = 15 ____(2)

solving (1) and (2)

A = 5 and C = 5

slope of the equipotential line

$\frac{dy}{dx} = \sqrt{3}$

and from equation (*)

0 = A dx + B dy

$\frac{dy}{dx} = -\frac{A}{B}$

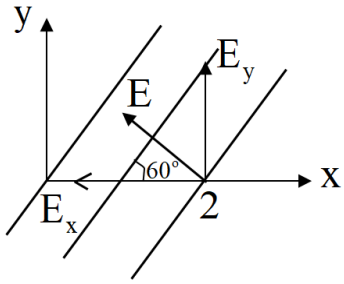
$\sqrt{3} = \frac{-5}{B} \Rightarrow B = \frac{-5}{\sqrt{3}}$

Hence

$V(x, y) = 5x - \frac{5}{\sqrt{3}}y + 5$

$V(x, y) = 5\left[x - \frac{y}{\sqrt{3}}\right] + 5$

OR



As we know

$\Delta V = -E_X \Delta x - E_X \Delta y + C$

here $E_X = -\frac{\Delta V}{\Delta x} = -5$

$E_Y = -\frac{\Delta V}{\Delta y} = +\frac{5}{\sqrt{3}}$

$V = +5x - \frac{5}{\sqrt{3}}y + C$

put x = 0, y=0, V = C = 5 Volt

$V = 5\left(x - \frac{y}{\sqrt{3}}\right) + 5$

35. Ans (3)

$F_m = F_e$

$qvB = qE$

$E = vB$

$V = Ed = vBd$

If B increases then V increases

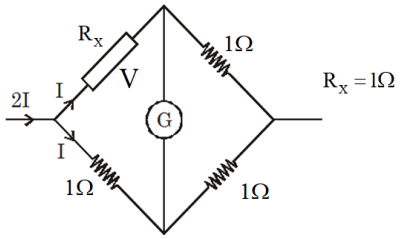
36. Ans (4)

$E_x = \frac{-\Delta V}{\Delta x}$

$E_y = \frac{-\Delta V}{\Delta y}$

$E_z = \frac{-\Delta V}{\Delta z}$

37. Ans (1)



$V = iR = I\,(1) = I$

$I = \frac{(I)^4}{125} \Rightarrow I^3 = 125$

$I = 5$ A

$i = 2I = 10$ A

38. Ans (1)

$I = neAV_d$

$V_d = \frac{I}{enA}$

$n = \frac{\rho}{M} \times N_A$; ρ is density

$V_d = \frac{IM}{eA\rho N_A}$

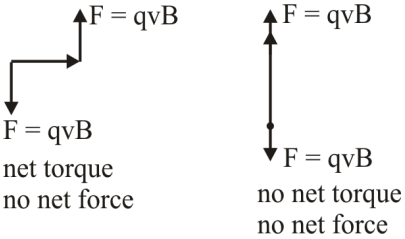
$V_d = \frac{10 \times 63.5 \times 10^{-3}}{1.6 \times 10^{-19} \times 0.05 \times 10^{-4} \times 8.9 \times 10^3 \times 6 \times 10^{23}}$

$V_d = \frac{63.5 \times 10 \times 10 \times 10^{-6}}{1.6 \times 5 \times 8.9 \times 6}$

$V_d \simeq 1.5 \times 10^{-4}$ m/s

39. Ans (3)

Since q_{net} on dipole = 0 $\Rightarrow$ no net force



The correct option is (3)

40. Ans (2)

$\phi = \vec{B}.\vec{A}$

$= BA\cos\theta$

$|\phi| = BA\,|\cos\theta|$

41. Ans (1)

Curie's Law states that $\chi_1 T_1 = \chi_2 T_2$

Given :

$\chi_1 = 1.2 \times 10^{-5}$

$T_1 = 300$ k

$\chi_2 = 1.8 \times 10^{-5}$

$T_2 =$

$\frac{\chi_1 T_1}{\chi_2} = \frac{(1.2 \times 10^{-5})(300)}{1.8 \times 10^{-5}} = \frac{1.2 \times 300}{1.8} = \frac{360}{1.8} = 200$K.

42. Ans (3)

Since V is leading I ($\theta < 90°$)

there must be L & R in circuit.

43. Ans (4)

$m = \frac{40}{40+40} = \frac{1}{2}$

$h_I = mh_0$

$h_I = 2$ mm

$dv = -m^2 du$

$dv = -1$ mm

hence area = 2 mm^2

44. Ans (4)

$I = I_0 + I_0 + 2\sqrt{I_0 I_0}\cos\phi,$

$\phi = \frac{2\pi}{\lambda}\Delta x = \frac{2\pi}{\lambda}\frac{\lambda}{4} = \frac{\pi}{2}$

so $I = 2I_0$

45. Ans (3)

For first minimum a sin $\theta = \lambda$

So $\frac{\lambda}{a} = \sin 37° = \frac{3}{5}$

Now for first secondary maximum a sin$\theta = \frac{3\lambda}{2}$

Therefore sin $\theta = \frac{3\lambda}{2a} = \frac{3}{2}\left(\frac{3}{5}\right) = \frac{9}{10}$

$\Rightarrow \theta = \sin^{-1}\left(\frac{9}{10}\right)$

46. Ans (2)

NCERT XIIth Haloalkane & haloarene Page No. 314

47. **Ans (4)**

NCERT XIIth Biomolecule, Intext Question

Page No. 10.1

48. **Ans (1)**

NCERT XIIth Amine Table 9.2

49. **Ans (2)**

NCERT XIIth Amine Example 9.5

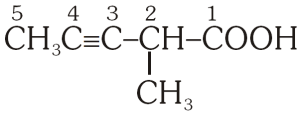
50. **Ans (1)**

NCERT XIth Part II, Problem 8.24

51. **Ans (1)**

NCERT XIIth haloalkane & Haloarene, Table 6.2

52. **Ans (2)**



2-Methyl pent-3-ynoic acid

53. **Ans (1)**

Conformational changes occur due to free rotation around the C − C sigma bond. During this rotation, bond lengths and bond angles remain constant. However, the dihedral angle (the angle between H atoms on adjacent carbons) changes. In the eclipsed conformation, electron cloud repulsion is maximized, leading to increased torsional strain. In the staggered conformation, this strain is minimized.

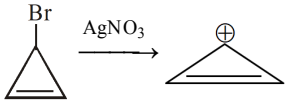
Final Answer: Option (1)

54. **Ans (1)**

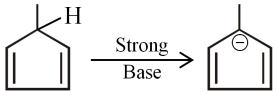
Solution: The pKa value is a measure of acidic strength. A low pKa value corresponds to a strong acid. The order of acidity is determined by the effects of the substituents. The correct match is A-Q, B-R, C-S, D-P. **Calculation**: N/A

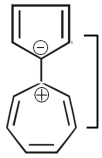
Final Answer : 1

55. **Ans (3)**

(A)  aromatic.

(B) Basic strength order ∝ Tendancy to donate L.P.

(C)  aromatic

(D)  aromatic

Correct statements are :- A, B, C, D

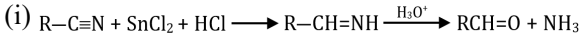
56. **Ans (2)**

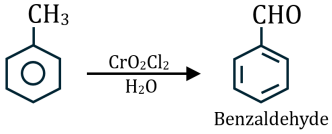
NCERT-XII, Part-II ; Page-330

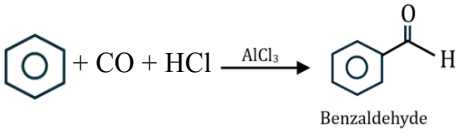
57. **Ans (4)**

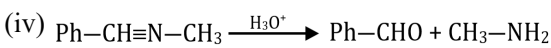
Reaction involves the formation of 2°free radical

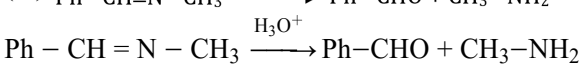
58. **Ans (4)**

(i) 

(ii) 

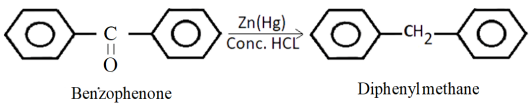
(iii) 

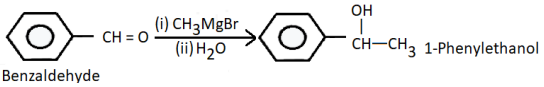
(iv) 

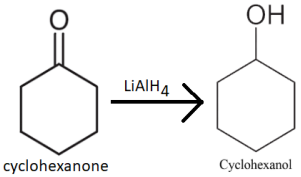


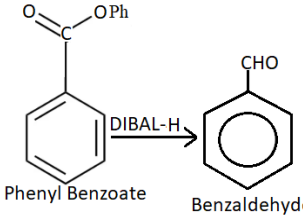
Final Answer : Option (4)

59. **Ans (2)**

1. 

2. 

3. 

4. 

The correct match is i-c, ii-d, iii-a, iv-b

Final Answer : option (2)

60. Ans (2)

Solution :

- (A) Chloroform and aniline → separated by distillation (P).
- (B) Glycerol + spent lye. → separated by distillation under reduced pressure (S) since glycerol decomposes at high temperature.
- (C) Aniline + water - steam → steam distillation (R) because aniline is steam volatile.
- (D) Amino acids → separated by Chromatography (Q).

Correct match is A–P, B–S, C–R, D–Q

Final Answer: Option (2)

61. Ans (2)

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62. Ans (3)

$$r = -\frac{d[R]}{dt} = +\frac{d[P]}{dt} = k[R]^1$$

$$\frac{d[R]}{dt} = -k[R]\,; \quad \frac{d[P]}{dt} = k[R]_0\, e^{-kt}$$

$$t_{1/2} = \frac{\ln 2}{k}$$

63. Ans (2)

5 molal solution = 5 moles of B in

1000 g of A

$$n_B = 5\ ;\ n_A = \frac{1000}{50} = 20$$

$$y_A = \frac{P_A^o X_A}{P_S}$$

$$= \frac{105\left(\frac{20}{25}\right)}{100}$$

$$= \frac{105}{125} = \frac{21}{25}$$

$$y_B = 1 - y_A$$

$$= \frac{4}{25}$$

64. Ans (4)

• For single-electron species (like Hydrogen), subshell energy depends entirely on the principal quantum number (n), meaning $E_{2s} = E_{2p}$.

• for multi-electron system (like Li and Na), the energy level depends on both n and ℓ due to electron shielding and nuclear pull. As the nuclear charge (Z) increases, the electrostatic attraction increases, causing the orbital energy levels to drop significantly down.

• Therefore, the true stability/energy trend is :

$E_{2s(H)} > E_{2s(Li)} > E_{2s(Na)}$.

Option 4 gives an inverted incorrect sequence.

65. Ans (3)

Using the boiling point elevation formula :

$\Delta T_b = k_b \times m \Rightarrow 0.5 = 5 \times m \Rightarrow m = 0.1$ mol/kg

This translates to 0.1 moles of solute in 1000 g of solvent.

The expression for the Relative Lowering of Vapour Pressure (RLVP) for a dilute solution is :

$RLVP = n_{solute} / n_{solvent} = m/(1000/M_{w(solvent)})$

$RLVP = 0.1/(1000/300) = 0.1 \times (300/1000) = 3 \times 10^{-2}$

Matching this with the given term $x \times 10^{-2}$, we find x = 3.

66. Ans (1)

Using Faraday's Laws, the required charge is evaluated by calculating the absolute equivalence change :

(A) Reduction to 1 mole of Cr^{3+} ion : The half-reaction involves a net gain of 3 electrons ($Cr^{6+} + 3e^- \longrightarrow Cr^{3+}$). Hence, charge = 1 × 3 = 3F (IV).

(B) 2 moles of $Cl_2(g)$ from molten $CaCl_2$: Each mole of Cl_2 requires 2 Faradays ($2Cl^- \longrightarrow Cl_2 + 2e^-$). For 2 moles : 2 × 2 = 4F (III).

(C) Electrolysis of 108 g of H_2O : Total moles of water = 108/18 = 6 moles. The decomposition of 1 mole of water shifts 2 equivalents of electrons ($H_2O \longrightarrow H_2 + \frac{1}{2}O_2$). Total charge = 6 × 2 = 12F (I).

(D) 54 g of Al from molten Al_2O_3 : Total moles of Al = 54 / 27 = 2 moles. Reduction of 1 mole of Al^{3+} requires 3 electrons. Total charge = 2 × 3 = 6F (II).

67. Ans (1)

Statement-I is true : Emulsions are inherently heterogeneous mixture of two or more liquids. If two liquids are completely miscible, they merge to form a uniform true solution rather than a colloidal dispersion.

Statement-II is true : In an oil-in-water (W/O) emulsion, water serves as the continuous outer dispersion phase. Therefore, adding external water seamlessly dilutes the continuous phase without breaking the structure.

68. Ans (2)

$$2SO_3 \rightleftharpoons 2SO_2 + O_2$$

initial	1	–	–
conc^n	1 – 0.8	0.8	0.4
at eq^m			

$$K_C = \frac{(0.8)^2(0.4)}{(0.2)^2}$$

$$= 6.4$$

$$\% \ \alpha = \frac{0.8}{1} \times 100 = 80 \ \%$$

69. Ans (4)

The standard reducing capacity of a chemical species is inversely related to its standard reduction potential (E°). The species with the lowest (most negative) reduction potential will show the maximum tendency to undergo oxidation, making it the strongest reducing agent. Comparing the given values : $E^{o}_{Cr^{3+}|Cr} = -0.74\,V$ is the minimum value. Hence, elemental chromium (Cr) acts as the strongest reducing agent.

70. Ans (4)

- **Statement A is correct :** Applying Hess's law: $\Delta H = \Delta H_2 - \Delta H_1 = -72 - (-69) = -3$ kJ/mol
- **Statement B is correct :** The value at infinite dilution represents the complete dissolution spectrum, matching the given asymptotic limit of −75 kJ/mol.
- **Statement C is correct :** The enthalpy of dilution structurally depends on the differential interaction profiles between the existing state concentrations and added volume thresholds.

71. Ans (2)

Let's evaluate the absolute atomic numbers using Avogadro's constant (N_A) :

(I) 22.4 L of H_2 at STP : Moles = 22.4 / 22.4 = 0.1 mol.

Atoms = $0.1 \times 2 \times N_A = 0.2\ N_A$.

(II) 6.84 g of sucrose ($C_{12}H_{22}O_{11}$, $M_W = 342$) :

Moles = 6.84 / 342 = 0.02 mol.

Atoms = $0.02 \times 45 \times N_A = 0.9\ N_A$.

(III) 0.1 moles of cyclohexane (C_6H_{12}) :

Atoms = $0.1 \times 18 \times N_A = 1.8\ N_A$.

The true decreasing order is therefore:

$1.8\ N_A$(III) > $0.9\ N_A$(II) > $0.2\ N_A$(I).

72. Ans (3)

For a weak acid solution, the classic hydronium equilibrium relation is :

$$pH = \frac{1}{2}(pK_a - \log C)$$

Given values : pH = 5 and concentration C = 0.1 M = 10^{-1}M.

$$5 = \frac{1}{2}(pK_a - \log 10^{-1})$$

$$10 = pK_a - (-1) \Rightarrow 10 = pK_a + 1 \Rightarrow pK_a = 9$$

73. Ans (1)

Using the ion-electron half-reaction method in an acidic medium :

- Oxidation half : $SO_3^{2-} \rightarrow SO_4^{2-} + 2e^-$ (Oxidation state increases by 2)
- Reduction half : $Cr_2O_7^{2-} + 6e^- \rightarrow 2Cr^{3+}$ (Oxidation state drops by 6 total)

Multiplying the oxidation step by 3 to balance the net electron exchange gives the stoichiometric coefficients : x = 1 and y = 3. Balancing the net remaining charge with protons yields z = 8 on the reactant side.

74. Ans (1)

$$2H^+ \longrightarrow 2H^+$$

$$0.1M \quad 0.01\ M$$

$$E_1 = 0 - \frac{0.0591}{2}\log\frac{10^{-4}}{10^{-2}}$$

$$= +0.0591\ V$$

$$E_2 = 0 - \frac{0.0591}{2}\log\frac{10^{-6}}{10^{-2}}$$

$$= +\frac{0.0591}{2} \times 4$$

$$= 2 \times 0.0591\ V$$

$$\Delta E = 0.0591\ V$$

75. Ans (1)

$\Delta U = W$

$nC_{v,m}\Delta T = W$

$2 \times \frac{3R}{2}(300 - 400) = W$

$W = -300\ R$

76. Ans (2)

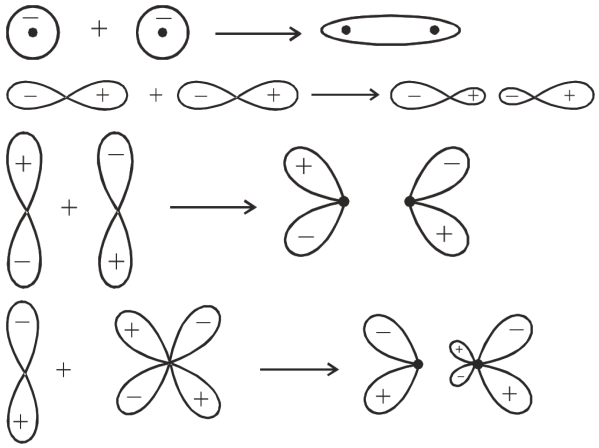
Question Explanation:

Which of the following molecular orbital formation is correct.

Concept:

This question is based on overlapping

Solution:



Final Answer: The correct option is (2).

77. Ans (3)

Explain Question:

Identify which is not related to H-bonding.

Concept:

Hydrogen bonding.

Solution:

Boiling point of hydrocarbon is greater than that of respective fluorocarbon due to more polarisability factor which is not associated to hydrogen bonding while rest properties are associated to the hydrogen bonding.

Final Answer: Option (3)

78. Ans (4)

Explain the question: Identify the anions that liberate a gas with dilute H_2SO_4 which turns baryta water turbid.

Concept: Qualitative analysis of acid radicals and precipitation reactions.

Solution

We analyze the behavior of the given anions when treated with sulfuric acid (H_2SO_4):

- CO_3^{2-} (Carbonate): Reacts with dilute H_2SO_4 to liberate CO_2 gas. CO_2 reacts with baryta water ($Ba(OH)_2$) to form a white precipitate of barium carbonate ($BaCO_3$), which produces turbidity.
- S^{2-} (Sulfide): Liberates H_2S gas, which has a rotten egg smell and does not produce turbidity with baryta water.
- SO_3^{2-} (Sulfite): Reacts with dilute H_2SO_4 to liberate SO_2 gas. Like CO_2,SO_2 reacts with baryta water to form a white precipitate of barium sulfite ($BaSO_3$), producing turbidity.
- Concentrated H_2SO_4 Reaction: Both carbonates and sulfites continue to produce their respective colourless gases (CO_2 and SO_2) when treated with concentrated acid.
- **Final Answer**

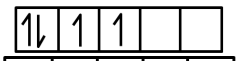
(CO_3^{2-}) and (SO_3^{2-})

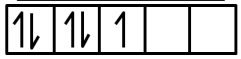
79. Ans (2)

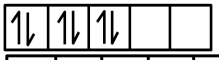
$[M(CN)_6]^{-3}$ $\mu = \sqrt{3}$

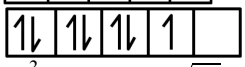
$x-6 = -3$ $\mu = 1.732$ BM $\Rightarrow n = 1$

$x = +3$ CN^- = SFL

$Mn^{+3} = d^4$  $\Rightarrow n = 2$ X

$Fe^{+3} = d^5$  $\Rightarrow n = 1$

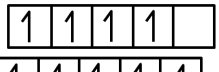
$Co^{+3} = d^6$  $\Rightarrow n = 0$ X

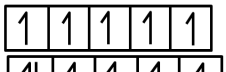
$Ni^{+3} = d^7$  $\Rightarrow n = 1$

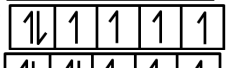
$[M(SCN)_6]^{-3}$ $\mu = \sqrt{35}$

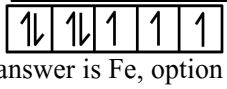
$x - 6 = -3$ $\mu = 5.91$ BM $\Rightarrow n = 5$

$x = +3$ SCN^- = WFL

$Mn^{+3} = d^4$  $\Rightarrow n = 4$ X

$Fe^{+3} = d^5$  $\Rightarrow n = 5$ ✓

$Co^{+3} = d^6$  $\Rightarrow n = 4$ X

$Ni^{+3} = d^7$  $\Rightarrow n = 3$ X

So, final answer is Fe, option 2

80. Ans (3)

The basic strength of the hydroxides decreases as we move from cerium to lutetium.

Also the lanthanoid hydroxides are more basic than $Al(OH)_3$ but less basic than $Ca(OH)_2$.

Both $B(OH)_3$ and $Si(OH)_4$ are acidic.

Hence choice (C) is correct while (A), (B) and (D) are incorrect.

81. Ans (3)

Question Explanation: $Cu(BO_2)_2$ (Blue) is reduced to $CuBO_2$(Colourless/Red opaque) in a luminous flame.

Concept: Borax Bead Test (Cu^{2+}).

Solution:

Statement I

(Luminous flame/Reducing flame): $Cu(BO_2)_2$ (Blue) is reduced to $CuBO_2$ or metallic Cu (Red opaque).

The colour change is from blue to red (opaque)

- or colourless. Stating the colour is green is **false**.
- **Statement II:** $Cu(BO_2)_2$ is CuII metaborate (Blue), not CuI metaborate.**False**.

Final Answer: Option

(3) Both Statement I and Statement II are false

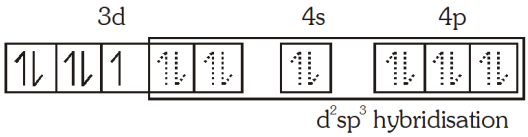
Statement- I :- In this case, the expected colour change is not to green but rather to colourless or a different fue depending on the oxidation state of copper

Statement II:- The bead does not twin green, instead it may twin clourless to red due to the reduction of cuporic ions to cuprous ion metalic copper.

82. Ans (4)

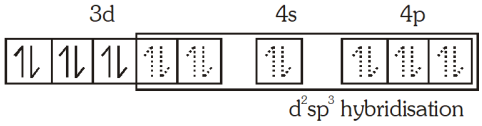
Solution

(A) $[\overset{III}{Fe}(en)_3]^{3+}$ -



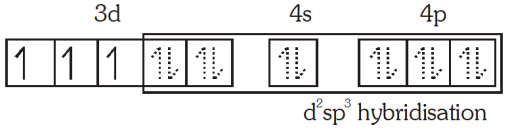
Number of unpaired electrons = 1; so paramagnetic

(B) $[\overset{III}{Co}(ox)_3]^{3-}$ -



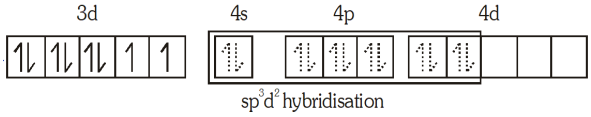
Number of unpaired electrons = 0; so diamagnetic.

(C) $[\overset{III}{Cr}(CN)_6]^{3-}$ -



Number of unpaired electrons = 3 ; so paramagnetic.

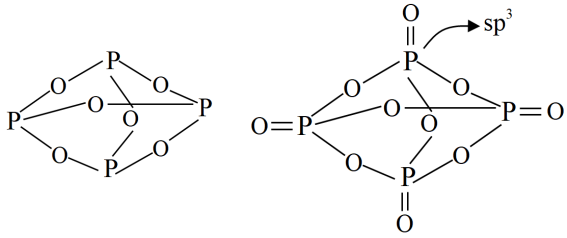
(D) $[\overset{II}{Ni}Cl_6]^{4-}$ -



Number of unpaired electrons = 2; so paramagnetic.

Suggested Solution (Allie)

83. Ans (3)



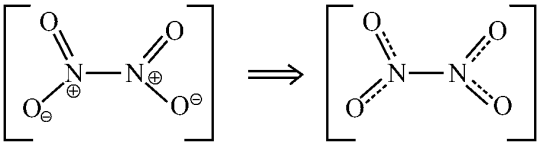
P_4O_6 (Anhydride of H_3PO_3) No π - bond

P_4O_{10} (Anhydride of H_3PO_4) Contains pπ-dπ bond

84. Ans (2)

N_2O_4 have equal resonating structure hence all

$-N-O$ bond length are identical to each other



Resonating hybrid

85. Ans (3)

In alkaline medium, $KMnO_4$ acts as a oxidizing agent.

86. Ans (4)

$Y b^{+2} \rightarrow Y b^{+3} + e^-$

$Y b^{+3}$ is more stable $\therefore$ Yb^{+2} is reducing agent.

87. Ans (4)

NCERT Pg#83, second last para

88. Ans (2)

Question Explanation: Match the salt reactions with their evolved gas properties.

Concept: Qualitative analysis of anions (Bromide, Sulphide, Nitrite, Nitrate).

Solution:

Reaction (Column-I)	Gas Produced	Key Properties (Column-II)
(P) NaBr + H_2SO_4	Br_2 vapours	(B) Brown colour
(Q) Na_2S + HCl	H_2S gas	(A) Colourless, (C) Rotten egg smell
(R) $NaNO_2$ + HCl	NO $\rightarrow$ NO_2	(B) Brown, (D) Paramagnetic
(S) $NaNO_3$ + H_2SO_4	NO_2 gas	(B) Brown, (D) Paramagnetic

Summary:

- P $\rightarrow$ B: Bromine fumes are brown.
- Q $\rightarrow$ A, C: H_2S is colourless with a rotten egg smell.
- R & S $\rightarrow$ B, D: NO_2 is brown and paramagnetic due to its odd electron.

Final Answer: Option (2)

89. Ans (3)

Explanation -

The question asks to identify the incorrect trend among the given options describing the properties (thermal stability, bond strength, reducing character, basicity) of Group 15 hydrides.

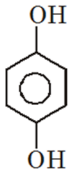
Concept - General Properties of Group - 15.

Solution -

The incorrect statement is regarding reducing character. Reducing ability increases down the group ($NH_3 < PH_3 < AsH_3 < SbH_3 < BiH_3$) because the stability of the hydrides decreases, making it easier to donate hydrogen. The other trends for thermal stability, bond dissociation enthalpy, and basicity decrease down the group.

Final Answer - Option (3)

90. Ans (2)



, NO_2^- : Polar

91. Ans (3)

NCERT Pg # 64

92. Ans (1)

NCERT Pg.# 64.

93. Ans (2)

NCERT-XI Pg. # 65, 69

Tomato, Lemon, Chinarose, Datura, Petunia

94. Ans (2)

NCERT XI, Pg. # 72

95. Ans (2)

NCERT XI Pg. # 74, 76

96. Ans (3)

NCERT XII Pg. # 19, 22, 23

97. Ans (1)

NCERT-XII, Pg. # 20

98. Ans (2)

NCERT XII, Pg. # 6,12,15,20

c and d are incorrect

99. **Ans (4)**
NCERT Pg. No. # 124

100. **Ans (1)**
NCERT Pg. # 125

101. **Ans (2)**
NCERT XI Page No.# 126, 127

102. **Ans (1)**
NCERT Page No. 98

103. **Ans (4)**
NCERT, Pg. # 95

104. **Ans (4)**
NCERT XII, Pg # 133

105. **Ans (4)**
NCERT XII pg no # 47

106. **Ans (3)**
NCERT pg.no #142

107. **Ans (3)**
NCERT Pg. # 140

108. **Ans (3)**
NCERT Pg. # 124-125

109. **Ans (2)**
NCERT Pg. # 121-122

110. **Ans (2)**
NCERT Pg. # 118

111. **Ans (2)**
NCERT Pg. # 36, 37, 38

112. **Ans (2)**
NCERT No. # 31

113. **Ans (3)**
NCERT Pg. # 33

114. **Ans (2)**
NCERT Pg. # 43

115. **Ans (1)**
NCERT Pg.#194.

116. **Ans (2)**
NCERT Pg. # 199

117. **Ans (1)**
NCERT Pg.# 185

118. **Ans (3)**
NCERT Pg. # 184

119. **Ans (1)**
NCERT Page No. #242

120. **Ans (2)**
NCERT Pg. # 242, 243

121. **Ans (3)**
NCERT Pg. # 239

122. **Ans (3)**
NCERT XII Pg. # 232

123. **Ans (3)**
NCERT XI Pg. # 237

124. **Ans (1)**
NCERT XI, Pg # 208

125. **Ans (3)**
NCERT, Pg. # 208

126. **Ans (4)**
NCERT Pg # 225

127. **Ans (1)**
NCERT Pg.#103 (old).

128. **Ans (3)**
NCERT Pg.# Module.

129. **Ans (1)**
NCERT Pg. No. # 83

130. **Ans (4)**
NCERT-XII, Pg. No. 51

131. **Ans (3)**
NCERT, Pg. # 49

132. **Ans (1)**
NCERT, Pg. # 49

133. **Ans (3)**
NCERT XI Pg.#40-43.

134. **Ans (3)**
NCERT-XII, Pg. # 199

135. **Ans (1)**
NCERT-XII, Pg # 233(E), 255(H)

136. **Ans (3)**
NCERT-XII, Pg. # 197, 198

137. **Ans (3)**
Question Explanation : Which of the following is correctly matched regarding decomposition.
Concept : Decomposition.
Solution :
• Decomposition : The breakdown of organic matter into simpler forms.
• Fragmentation : Physical breaking down of detritus into smaller particles.
• Humification : Formation of dark colored amorphous substances like humus.
• Leaching : Movement of soluble substances like nutrients down through soil layers.
• Minerlisation : Release of inorganic nutrients from organic matter by microbes.
Final Answer : Option (3)

138. **Ans (1)**
Explain Question: The question is asking for the term used to describe how different species are arranged at various vertical levels within a biotic community, such as trees, shrubs, and herbs forming distinct layers.
Concept: Stratification.
Solution:
Vertical distribution of different species occupying various levels in a biotic community is termed stratification.
For example:
- Trees occupy the top layer.
- Shrubs form the second layer.
- Herbs and grasses occupy the bottom layer.
Final answer: Option (1)

139. **Ans (2)**
NCERT Pg. # 207

140. **Ans (3)**
NCERT-XII, Pg. # 220

141. **Ans (4)**
NCERT Page No. # 263

142. **Ans (4)**
Correct Answer (4)

143. **Ans (2)**
NCERT Pg. # 245, 246

144. **Ans (3)**
NCERT-XII, Pg. # 222, 223, 225

145. **Ans (1)**
Correct Answer: (1)

146. **Ans (3)**
NCERT Page No. 65-71

147. **Ans (3)**
NCERT Pg.#60,61,62,69 (E), 67,68,69,77 (H)

148. **Ans (2)**
NCERT Page No. : 57

149. **Ans (2)**
Both Parents affected but progeny unaffected so autosomal dominant.

150. **Ans (1)**
NCERT Pg#92

151. **Ans (1)**
NCERT Pg # 100

152. **Ans (1)**
NCERT Pg. # 96

153. **Ans (4)**
Question explanation: The question asks which pairing between a type of RNA polymerase (or related factor) and its product/function is incorrect.
Concept :
Transcription
Explanation :
Sigma factor: In prokaryotes, sigma factor helps RNA polymerase recognize the promoter; it is not a termination factor. Termination factors are separate proteins (like Rho in bacteria).
Final answer : 4

154. **Ans (2)**
Phosphodiester bond is formed between 3'−OH and 5'− P

155. Ans (2)

NCERT Pg. # 89, 90

156. Ans (2)

NCERT Pg. # 170

157. Ans (1)

NCERT Pg. # 172

158. Ans (4)

NCERT XII Page No. # 183

Recognition sequence is palindromic

159. Ans (3)

NCERT-XII, Pg # 183-184

160. Ans (1)

NCERT-XII, Pg. # 179 (E), 196 (H)

161. Ans (2)

NCERT Pg # 180, 183

162. Ans (3)

Explain Question : The question asks us to match four biological products (Acetic acid, Cyclosporin A, Statin, Streptokinase) with their source organisms and their uses from given columns. We must identify the combination where all matches are correct.

Concept : This question is based on Acetic acid ⟶ Found in vinegar, preservative.

Cyclosporin A ⟶ Produced by fungus Trichoderma polysporum, acts as immunosuppressive agent (prevents organ transplant rejection).

Statin ⟶ Produced by yeast Monascus purpureus, lowers blood cholesterol.

Streptokinase ⟶ Bacterial enzyme, used as clot buster (dissolves blood clots).

Explanation :

(I) Acetic acid: Source – vinegar (d), Use – preservative (iv).

(II) Cyclosporin A: Source – Trichoderma (b), Use – immunosuppressive agent (iii).

(III) Statin: Source – yeast (a), Use – blood cholesterol lowering (ii).

(IV) Streptokinase: Source – bacterial product (c), Use – clot buster (i).

This matching perfectly corresponds to:

I – d – iv, II – b – iii, III – a – ii, IV – c – i.

Final answer : 3

NCERT Pg. # 153

163. Ans (4)

Both 1 and 3

NCERT Pg. # 157

164. Ans (1)

Question Explanation: we need to identify which stage of sewage treatment physically removes solid particles.

Concept: sewage treatment

Solution: The primary treatment of sewage involves physical processes like filtration and sedimentation to remove floating and suspended particles.

Final answer: 1

NCERT Pg. # 154

165. Ans (3)

NCERT (XI) Pg. # 4

166. Ans (2)

NCERT XI, Page # 14,15

167. Ans (1)

NCERT XIth Pg.#17

168. Ans (3)

NCERT-XIth, Pg.# 29

169. Ans (4)

NCERT-XI, Pg. # 26

170. Ans (2)

NCERT-XI, Pg.# 27

171. Ans (2)

NCERT-XI-Pg. # 110

172. Ans (2)

NCERT-XI, Pg. # 109, 110

Question Explanation

This question asks you to match different proteins with their correct descriptions or functions.

Concept

Biomolecules ⟶ Proteins ⟶ Types and functions of proteins

Explanation

Collagen is the most abundant protein in animals, GLUT-4 transports glucose into cells, Trypsin is a digestive enzyme, and RuBisCO is the most abundant enzyme in the biosphere.

Final Answer

Option 2 — A–III, B–IV, C–I, D–II.

173. Ans (2)

NCERT-XI, Pg. # 104, 108, 109

The correct option is: **(2) A - a,c,d ; B - b,e,f**

Explanation:

- **A - (a) Acid-soluble pool:** Contains small molecules, such as **amino acids, sugars, nucleotides (d)**, with molecular weights ranging from **18 to 800 daltons (c)**.
- **B - (b) Acid-insoluble pool:** Contains larger molecules like **proteins, nucleic acids, polysaccharides (f)**, with molecular weights **greater than 1000 daltons (e)**.

Hence, option (2) is correct.

174. Ans (4)

NCERT-XI, Pg. # 117

Question Explanation: The question asks which statement correctly describes enzyme inhibition.

Concept: Enzyme activity

Solution:

Competitive inhibitor resembles the substrate and competes for the active site of the enzyme.In competitive inhibition, Km increases (lower affinity), but Vmax remains unchanged.Malonate inhibiting succinate dehydrogenase is a classic competitive, not non-competitive, example.Non-competitive inhibitors do NOT resemble the substrate. So only statement that is Option 4

Final answer: Option 4

175. Ans (1)

NCERT-XI, Pg. # 146, 147

176. Ans (4)

NCERT (XI) Pg. # 144

177. Ans (4)

NCERT-XI, Pg. # 157, 161, 162, 163

178. Ans (2)

NCERT-XI, Pg. # 156, 157, 158, 159

179. Ans (2)

NCERT-XI, Pg. # 174, 175

180. Ans (2)

NCERT-XI, Pg # 176, 177